

Project Phase 2

CMPT 276: Introduction to Software Engineering
Simon Fraser University

Group #15

Megan Chau	Student #301614135
Samantha Gan	Student #301614044
Krish Kaushik	Student #301628888
Tanveer Muntaseer	Student #301620006
Lucas Nguyen	Student #301629229

March 10, 2026

Contents

1	Milestones	1
1.1	Midway Milestone Goals	1
1.2	Final Milestone Goals	1
1.3	To Implement in Phase 3	2
2	Report	2
2.1	Approach	2
2.2	Modifications from Phase 1	2
2.3	Team Management	3
2.4	External Libraries	4
2.5	Challenges	4

1 Milestones

1.1 Midway Milestone Goals

Implementation

- Maven set up
- libGDX package setup
- WASD controls the character
- Camera tracking
- Start screen skeleton
- First area of the map
- Police Enemy
- Puddle Enemy

Design

- Points Counter
- Main character design and frames on and off the bike
- Enemy design (police and puddles) and animations
- Instructions Screen
- Settings Screen

Areas not Met

- Entire map was not completed as planned
- Design of assets was not completed
- Chose music and sound effects but were not implemented

1.2 Final Milestone Goals

Implementation

- Add the entire connecting map
- Points System
- Collision
- Sound effects and Music
- Delivery System
- Menus, Settings, and Pause
- Game Timer

Design

- Map 2 & 3
- Fish
- Orders
- Pause Screen
- End Screen

Areas not Met

- End screen not finalized – still using mockup version
- Cat mounting and dismounting bike
- Enemy positioning
- Optimize camera movement

1.3 To Implement in Phase 3

Due to time constraints, there were certain features we weren't able to implement during Phase 2. We will implement the following features during Phase 3:

Gameplay and Feature Improvements

- Have the main character mount and dismount a scooter (catOnBike sprite in resources)
- Have the pizza orders show up on the bottom right of the screen as they come in (optimized delivery system)
- Safety from police when transitioning between maps
- Menu button on pause screen
- Improve randomized movement of police (avoid walking into barriers)

Design and Polishing Improvements

- Create a starting and ending cut scene of the cat mounting the bike
- UI improvements for settings screen and pause game screen
- Replace placeholder ending screen with a Loss vs Win screen (sound cues, text, etc.)
- Clear requirements for each level on the top right of the screen

2 Report

2.1 Approach

- We used agile-style development with short iterations to allow incremental building and adjustment as problems arose.
- implemented libGDX as our backbone library.
- Features were developed in small modules (player movement, enemies, scoring, deliveries) and then integrated into the main game.
- Team roles were split into Design, Front End, and Back End.
- We maintained meetings at least once per week where we coordinated, tested, and collaborated. This approach worked well with our short iteration-style development.

2.2 Modifications from Phase 1

- We originally planned for a dog police, but we ended up changing it to a human.
- Unlock the next part of the map with fish instead of deliveries.
- All point actions (Fishes, Deliveries, Enemy collision deductions) all translate to a single displayed 'Fish' score, instead of all being separate counters.

- Because we decided to use the libGDX framework after our initial planning stage, the structure of our code is completely different:
 - libGDX helped us cut down the number of classes/code needed. For example:
 - * The import, "import com.badlogic.gdx.graphics.OrthographicCamera" is libGDX's built-in camera class, making it easier for us to use the code for the camera movement
 - * Built in render(), show(), dispose(), etc. methods
 - We have separate classes for separate game screens, which we did not anticipate that we needed

Initial Classes (without libGDX)	Finalized Classes (with libGDX)
Title, Instructions, Gameplay, GameState, GameWorld, GameController, InputHandler, Direction, Town, CollectibleManager, CollisionManager, Sprite, Barrier, CatFood, Tilemap, MainCharacter, Enemy, NonMovingEnemy, MovingEnemy, Score, Delivery, DeliveryManager, DeliveryPoint, DeliveryStatus, TransportMode, Position	EndScreen, Fish, GameCamera, GameLauncher, GameScreen, House, Instructions Screen, Main, Player, PoliceEnemy, PuddleEnemy, SettingScreen, StartScreen, Timer

2.3 Team Management

- Assigned clear roles (front end, back end, design) that changed throughout the project based on circumstance and completion
- Divided tasks in each role based on the order of requirements so work could support other group members
- Ensured everyone always had something to work on
- Held weekly meetings to check progress

Front End	Megan, Krish, Samantha, Lucas
Back End	Lucas, Tanveer, Krish, Samantha
Design	Megan, Tanveer

2.4 External Libraries

For this project we used the external library: **libGDX**

- It is used in many well-known games.
- It is a widely used library with strong community support and documentation.
- It is versatile and relatively easy to implement and use.
- It provides the tools needed to handle our game's graphics, input, and overall functionality.

2.5 Challenges

1. Balancing midterms and coursework from other classes. We managed our time carefully, ensured everyone participated in weekly meetings, and communicated to find meeting times that worked for all group members.
2. Many of our initial commits were made using personal email addresses, which did not appear on the commit log and have skewed the commit history by hundreds of lines of code. After identifying this issue, we corrected it, and all subsequent commits were made using our SFU email addresses.